Pseudorandom Functions, Linear Codes, and Substitution-Permutation Networks

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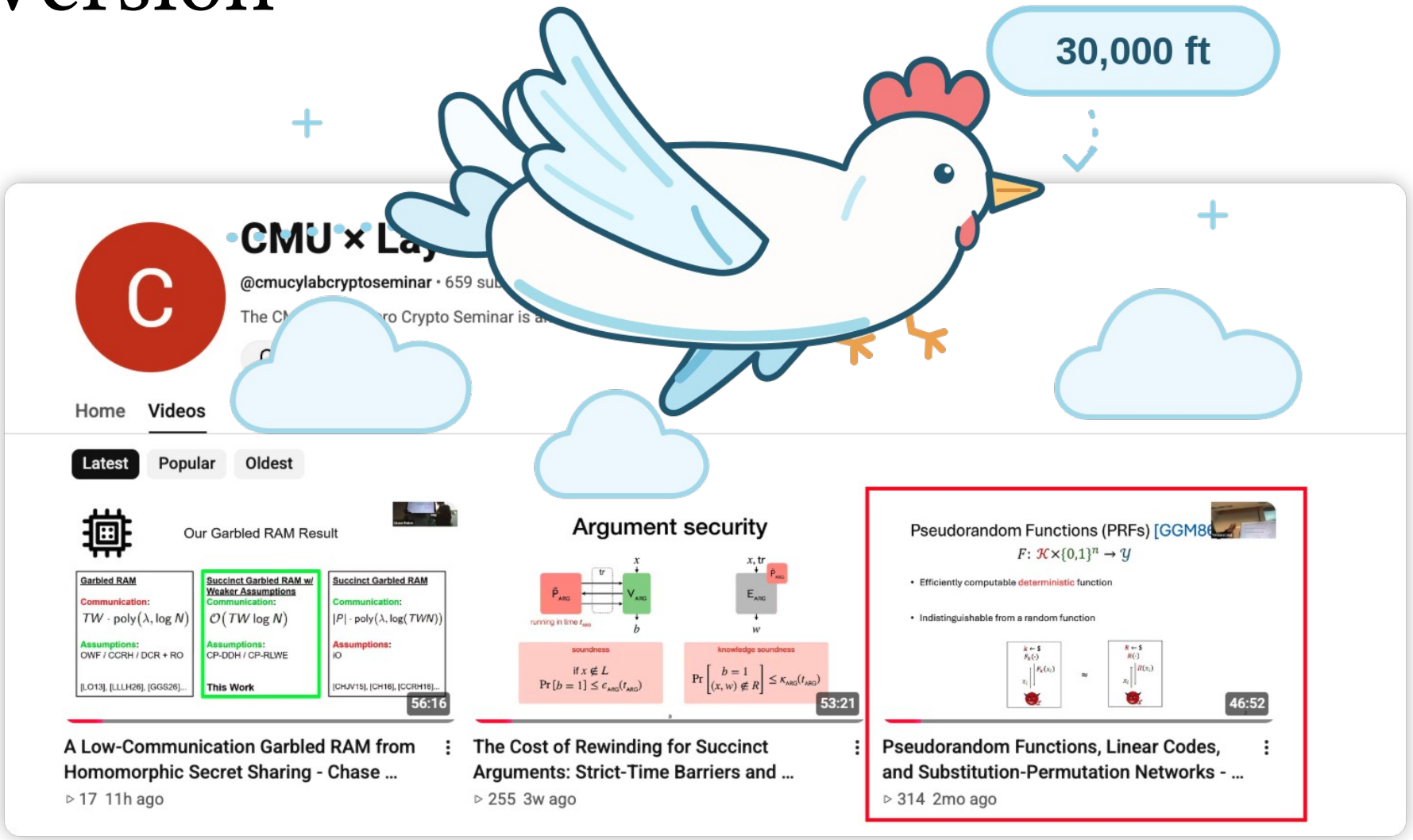
Joint work with Aayush Jain, Ilan Komargodski

[STOC'25] A New Approach for LPN-based Pseudorandom Functions: Low-Depth and Key-Homomorphic

[TCC'26] Logarithmic-Depth Pseudorandom Functions from Well-Founded Code-Based Assumptions

A Longer Version





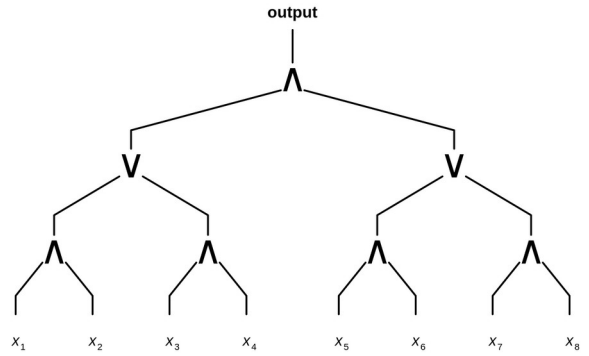
Pseudorandom Functions (PRFs) [GGM86]

$$F: \mathcal{K} \times \{0,1\}^n \rightarrow Y$$

- Efficiently computable deterministic function

- Indistinguishable from a random function

NC1 PRFs, good



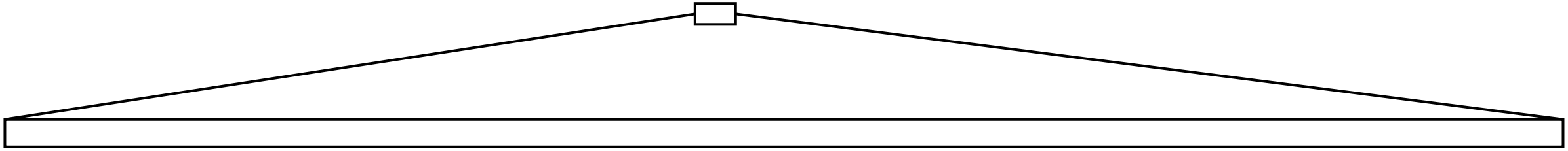
$O(\log n)$-depth Boolean circuits with bounded fan-in gates

- Optimality
  - Optimal size $O(n)$ comes for free [IKOS08]
  - The only bottleneck is achieving optimal depth $O(\log n)$

- Why depth?
  - High-end cryptographic primitives [ARS+15,MJSC16,BIP+18,BCG+20]
  - MPC [DI06,IKOS08,GVW12,BJKL21]
  - Circuit lower bounds [RR94,MV12]
  - Derandomization [NW88,Wil13]
  - Learning complexity [Val84,KV94]
  - Meta complexity [AN25]

Revisiting [GGM86]

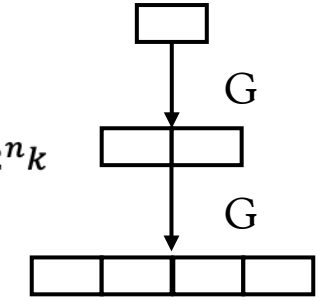
PRF is a PRG with exponential stretch

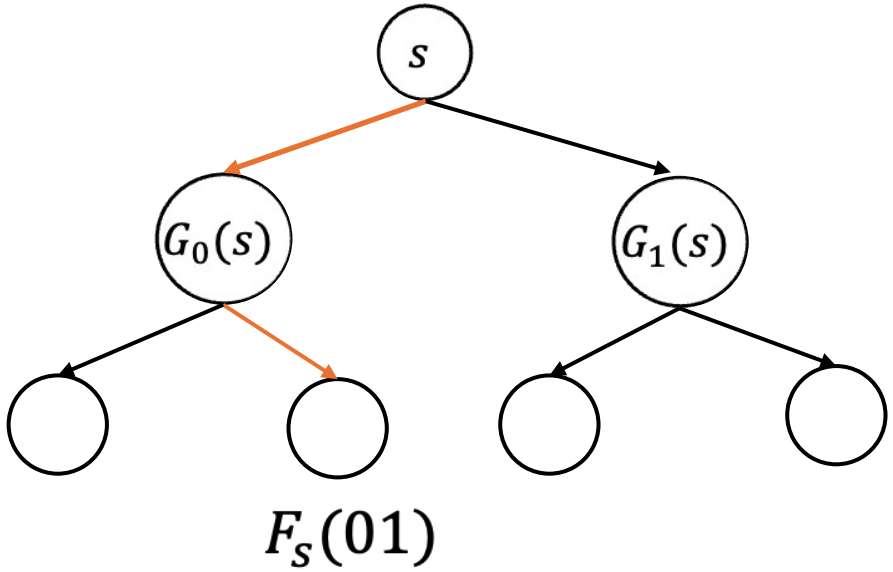


Repeat

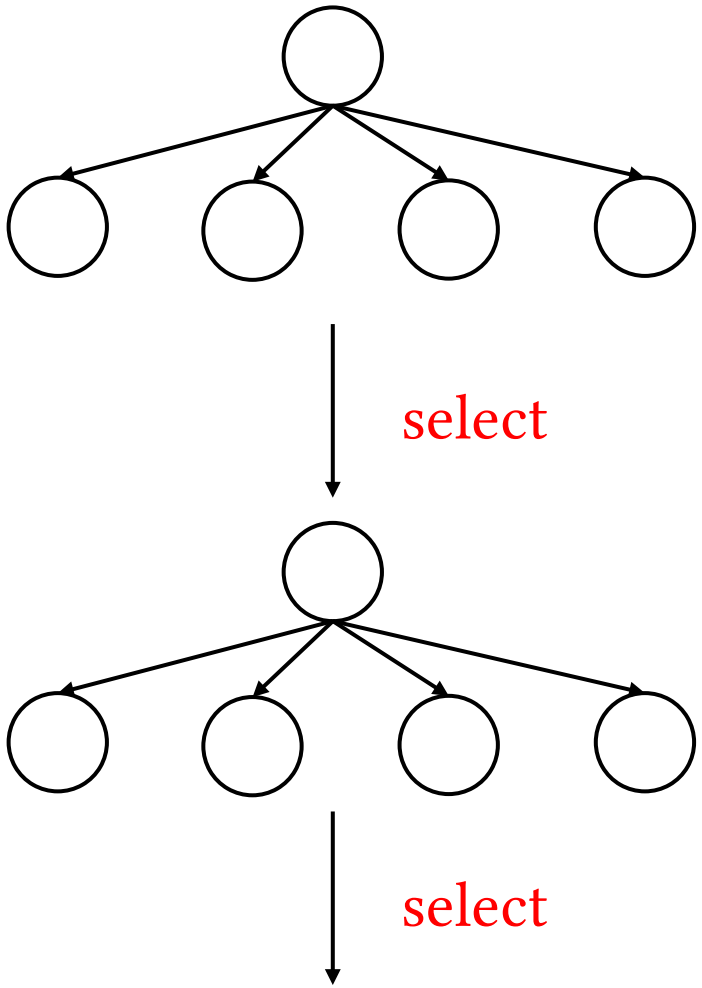
PRG $G: \{0,1\}^k \to \{0,1\}^{2k}$

$G^n: \{0,1\}^k \to \{0,1\}^{2^n k}$





NC1 PRF from PRG via GGM?



Poly-stretch PRG $G: \{0,1\}^{\lambda} \to \{0,1\}^{\lambda^2}$

Input domain $\mathcal{X} = \{0,1\}^{\omega(\log \lambda)}$

On $x = (x_1, \ldots, x_\ell), x_i \in [\lambda], \ell = \omega(1)$:

$$F_s(x) = G_{x_\ell} \circ \cdots \circ G_{x_1}(s)$$

- If G is in NC^0, i.e., constant depth
- Then the depth of PRF is $O(\ell) = O(\log n)$?
- $O(\ell + \ell \cdot \log n) = \omega(\log n)$, not in NC^1

NC1 PRF from PRG via ~~GGM?~~ Specific Assumptions

- PRFs in NC^1 from:
  - DDH [NR97]
  - Factoring [NRR02]
  - k-Linear [LW09]
  - Ring-LWE [BPR12]
  - Strong Goldreich's OWF [AR16]
  - Code-based, e.g., LPN?
    - Best known: [YS16] $NC^{1.01}$

- Direct PRFs
  - TC^0 [MV12]
  - $AC^0[2]$ [Ball et al. 26]

↘

Tomorrow morning!

Learning Parity with Noise (LPN

Cryptographic Primitives Based on Hard Learning Problems

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[3] Princeton University, Princeton, NJ 08544

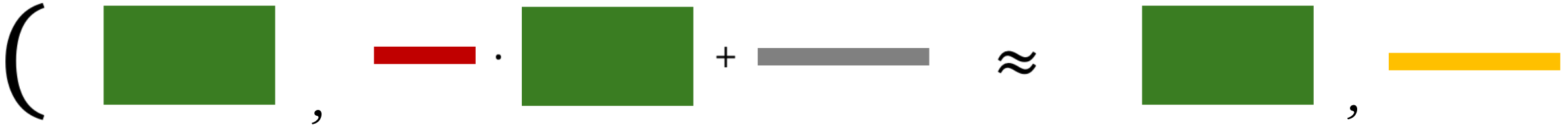
[BFKL93]

$$(A, sA + e) \approx (A, u)$$

- LPN: samples of the form $\langle a_i, s \rangle + e_i$ is random even given a_i

$$A \leftarrow \mathbb{Z}_2^{n \times m} \qquad s \leftarrow \mathbb{Z}_2^n \qquad e \leftarrow Ber_\mu^m$$

$$Ber_\mu = \begin{cases} 1 & \text{w.p. } \mu \\ 0 & \text{w.p. } 1-\mu \end{cases}$$



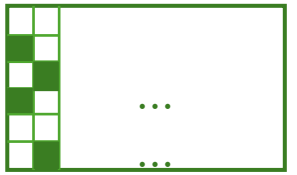
- Parameter: #sample $m(n)$, noise level $\mu(n) > 1/n$

LPN, good

- Plausibly post-quantum secure
  - Conservative
- Active area of research
- Equivalent to the search version
- Decoding random linear codes is hard

Our Results

Result 0: Weak PRFs in NC^1 from LPN [DJK'25]
Result 1: PRFs in NC^1 from Sparse-LPN* [DJK'26]
Result 2: PRFs in NC^1 from Ring-LPN* [DJK'26]
Result 3: PRFs in NC^1 from quasi-poly LPN [DJK'26]



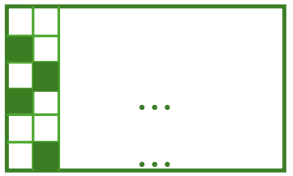
- LPN noise level: $\mu = n^{-o(1)}$ noise $(n^{-0.1} < \mu < 0.1)$
- Our parameter regime is not known to imply PKE/CRHF

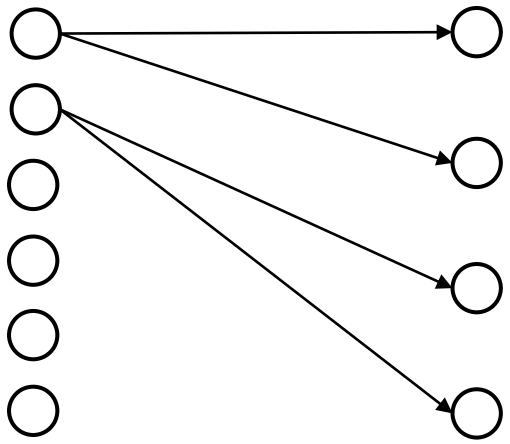
Sparse LPN: The Expander Graph Perspective

$$sA + e, A = (A_1, A_2, \dots, A_n) \in \{0,1\}^{n \times n^2}$$

Expander:

$$\phi_A : [n^2] \to [n]^d$$





$$\forall S \subseteq [n^2], s.t. |S| \leq N, \text{ it holds that}$$

$$|\phi(S)| > \frac{1}{2} \cdot d|S|$$

This will imply that

any N columns of A is linearly independent

- Random expander

Sparse LPN w.r.t. a *sublinear*-depth expander

$$sA + e, A = (A_1, A_2, \dots, A_n) \in \{0,1\}^{n \times n^2}$$

$$\phi_A : [n^2] \to [n]^d$$

- Random expander
  - *linear* depth, i.e., $O(\log n)$
- What is required
  - sublinear depth, i.e., $o(\log n)$

- [Alekhnovich03]: Conjectured that SparseLPN holds w.r.t. any expander (⋆)

  [MST'19][BLRS'26][HS'26]
  - The attack works for specific choice of the expander.
  - (⋆) is false

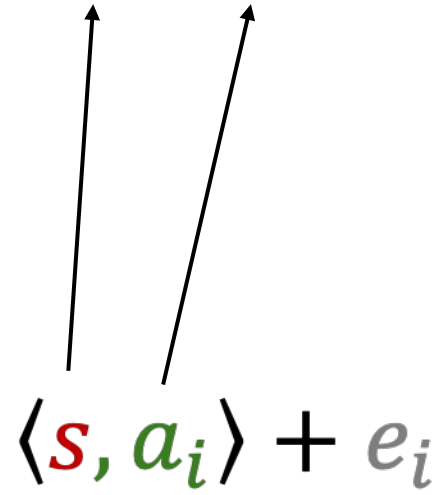
Remark 1 It can be shown that if the matrix A_n happens to be "degenerate" (for example, contains two equal rows, which occurs with probability $1/n$) then one can distinguish vectors b_1 and b_2 with probability roughly $1/n$. This gives an algorithm that distinguishes (A, b_1) and (A, b_2) with success $1/n^2$. We believe that no algorithm can do substantially better than this bound, and if A_n is an expander (which occurs with probability $1 - O(1/n)$) then the distributions of b_1 and b_2 are indistinguishable. Thus, we could specify in Problem 2 that A is chosen uniformly from the set of good expanders and assume its $1/n^{\Omega(1)}$-intractability (but this would sacrifice the property of being samplable).

Weak PRFs from LPN [DJK25]

Weak PRF:

$$\left(r_i, F_k(r_i)\right) \approx (r_i, u_i)$$

$$F\colon \mathcal{K} \times \mathcal{X} \to \mathcal{Y}$$



$$\langle s, a_i \rangle + e_i$$

Pseudorandom Functions (PRFs) [GGM86]

$$F: \mathcal{K} \times \{0,1\}^n \to \mathcal{Y}$$

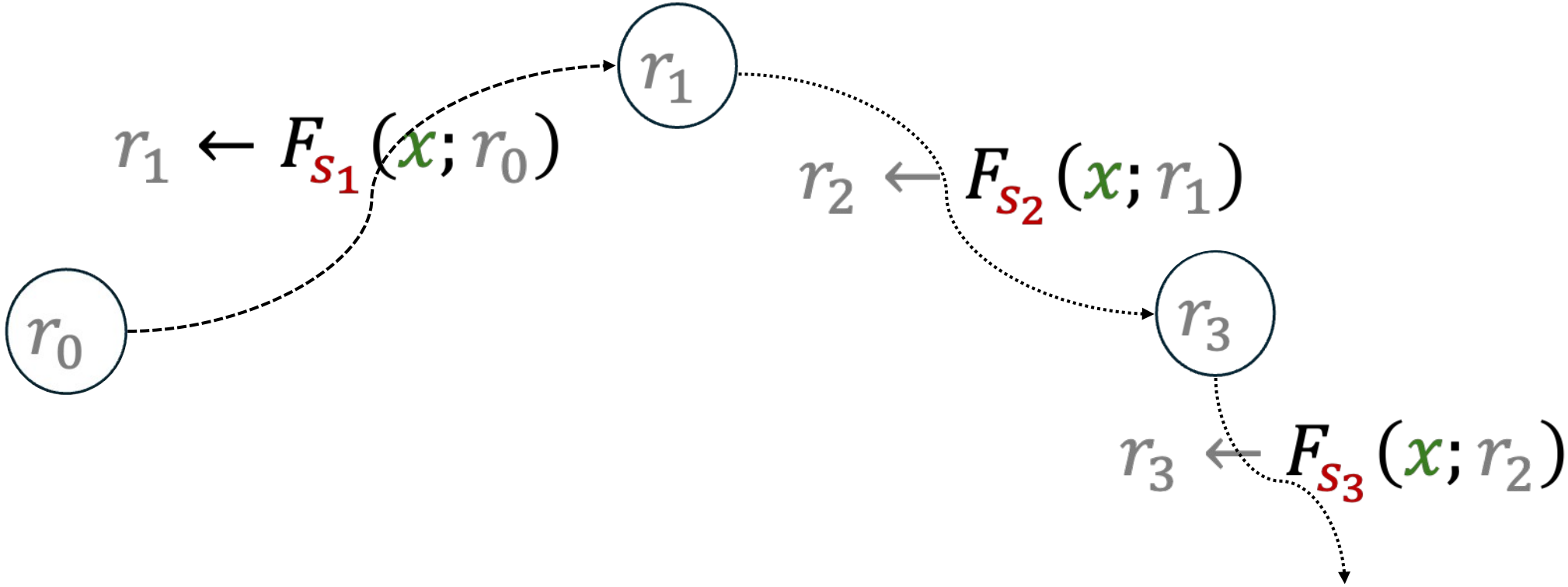
- Efficiently computable ~~deterministic~~ function
  randomized

$F(x; r)$

- Indistinguishable from a random function

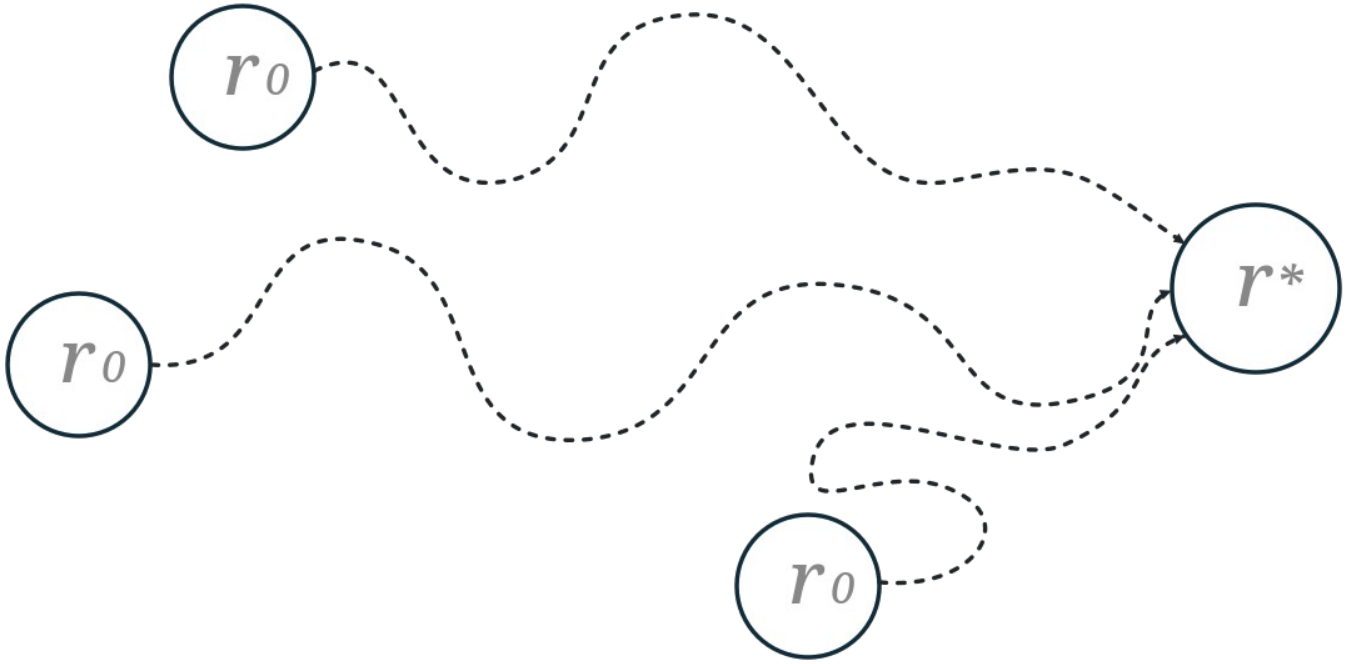
A Pseudo-Random Walk

Fix $x, (s_1, s_2, s_3, \dots)$



A Pseudo-Random Walk

Fix x, $(s_1, s_2, s_3, \ldots)$



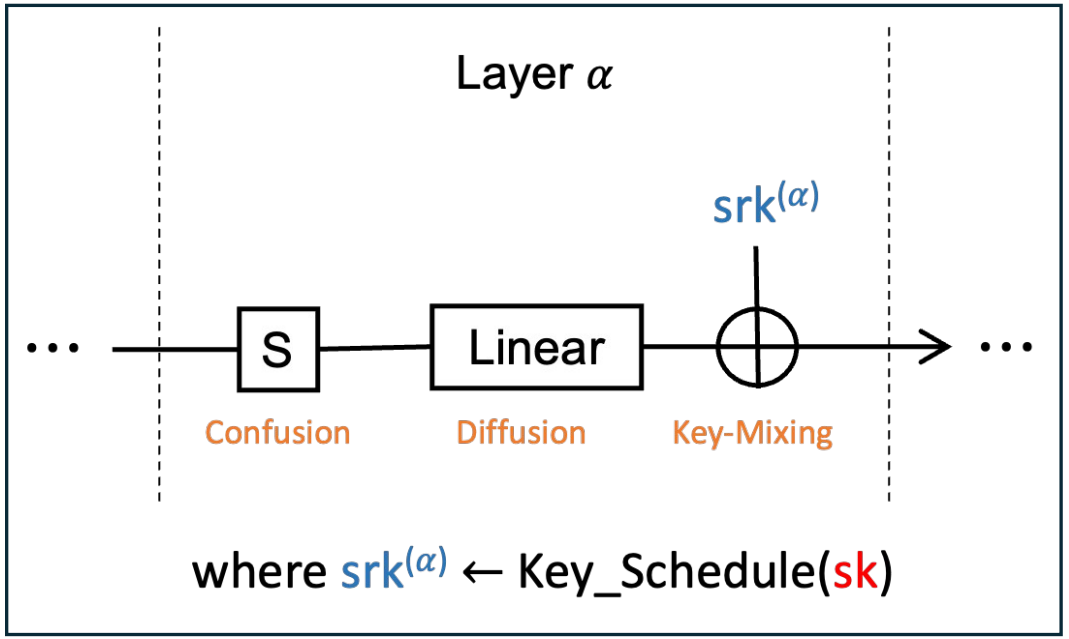
Collision of Bernoulli Random Variables

For any distribution (e, e') support on $\mathbb{Z}_2^m \times \mathbb{Z}_2^m$
If each marginal e, e' is distributed as Ber_μ^m,
Then

$$\Pr[e \neq e'] \leq 2m\mu$$

If $m \ll \frac{1}{\mu}$, (i.e., $m \ll n$) then we have good probability that they will collide

Substitution-Permutation Networks [Shannon49]



- Key Schedule: $\text{srk}^{(\alpha)} \leftarrow \langle s, A_x \rangle$
- S := S-Ber = $\prod x_i$
- Linear: A'_x

S-Box



$x \mapsto x^{-1}$ (AES)
$x \mapsto x^3$ (MiMC)
...



Sample Bernoulli noise
$x \mapsto \prod x_i$

Why PRFs from LPN?

- Curiosity
- First Key-Homomorphic NC1 PRFs
  - The construction has the form $sA + e$
  - $F_{k_1+k_2}(x) \approx F_{k_1}(x) + F_{k_2}(x)$
  - Caveat: For KH, only support a small input domain $\{0,1\}^{\omega(\log \lambda)}$
- Puncturable NC1 PRFs
- Substitution-Permutation Networks

Open Problem:

- More explicit constructions of sublinear-depth expanders?
- PRFs in NC1 from LPN?
  - Yes

Follow-up

Pseudorandom Functions in NC^1 from LWE/LPN/CDH (Or: How to Build Low-Depth PRFs, Generically)

Youlong Ding* Aayush Jain† Ilan Komargodski‡

Abstract

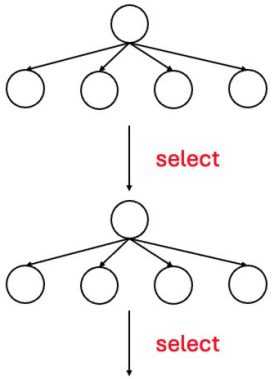
We obtain new NC^1-computable pseudorandom functions (PRFs) from various standard assumptions, resolving several long-standing open problems. Concretely, for the first time, we obtain NC^1-computable PRFs:

- from the **Learning With Errors (LWE)** assumption with a polynomial modulus-to-noise ratio, improving upon prior low-depth constructions that required Ring-LWE with super-polynomial ratios [Banerjee-Peikert-Rosen, EUROCRYPT 2012];
- from the **Learning Parity with Noise (LPN)** assumption, removing the need for structured LPN variants [Boyle et al., FOCS 2020], [Ding-Jain-Komargodski, STOC 2025], [Ding-Jain-Komargodski, TCC 2026];
- from the **Computational Diffie-Hellman (CDH)** assumption; prior works relied on the stronger Decisional Diffie-Hellman (DDH) or generalized Diffie-Hellman (GDH) assumptions [Naor-Reingold, FOCS '97, J. ACM '04].

Technically, our results follow from a new generic transformation that converts weak PRFs computable in depth $d(n) = \Omega(\log n)$ to (strong) PRFs computable in depth $O(d(n))$. Roughly speaking, our approach combines *tapering* the internal state of the classical GGM tree (so that the per-level depth decreases geometrically) with a two-stage security proof that transitions from a computational to an information-theoretic argument.

STOC '27 under (everyone's) review

NC1 PRF from PRG via GGM?



Poly-stretch PRG $G: \{0,1\}^{\lambda} \to \{0,1\}^{\lambda^2}$
Input domain $\mathcal{X} = \{0,1\}^{\omega(\log \lambda)}$
On $x = (x_1, \ldots, x_\ell), x_i \in [\lambda], \ell = \omega(1)$:

$$F_s(x) = G_{x_\ell} \circ \cdots \circ G_{x_1}(s)$$

- If G is in NC^0, i.e., constant depth
- Then the depth of PRF is $O(\ell) = O(\log n)$?
- $O(\ell + \ell \cdot \log n) = \omega(\log n)$, not in NC^1

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